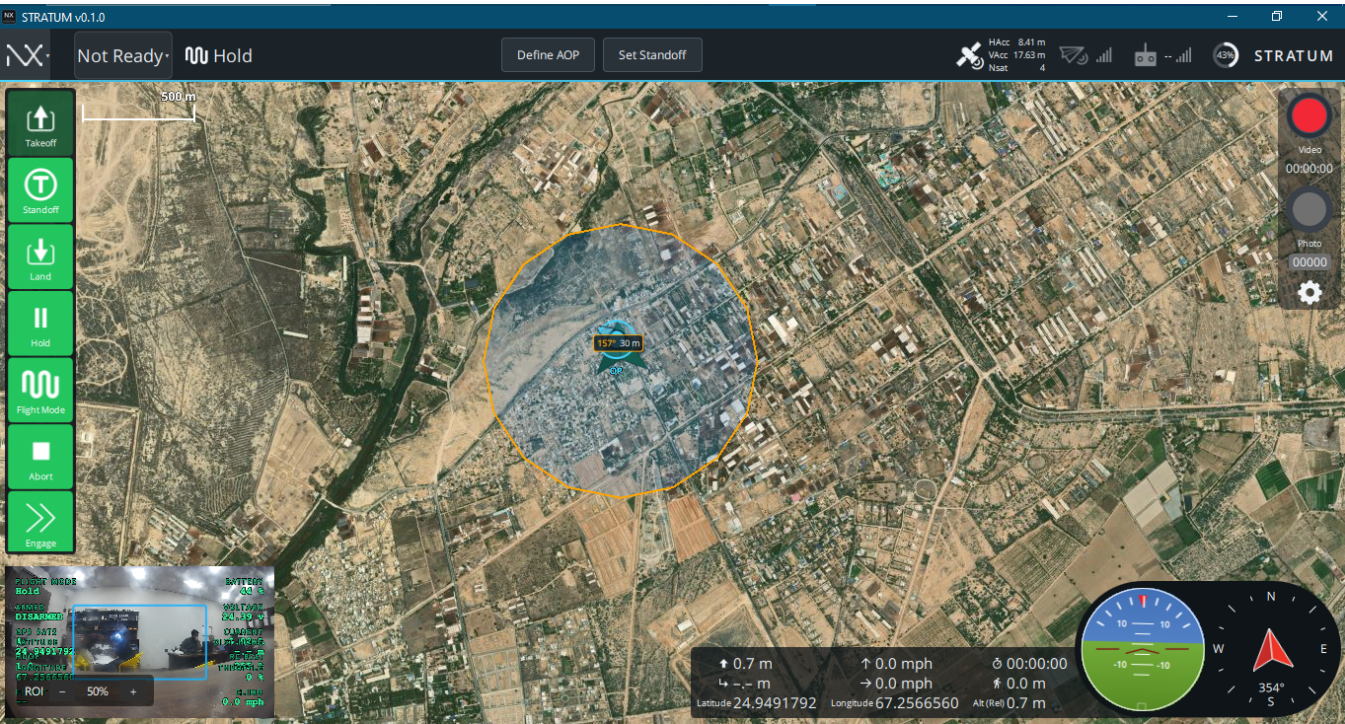


STRATUM Ground Station -- User Guide

This guide walks you through everything you see on the STRATUM screen while flying: the top ribbon, the command buttons on the left, the map, the video window, the right-hand camera controls, and the flight instruments along the bottom. Each section pairs a screenshot with a plain-English explanation of what every button and readout does.

Screenshots in this guide live in the `UI/` folder next to this document.

1. The main screen at a glance



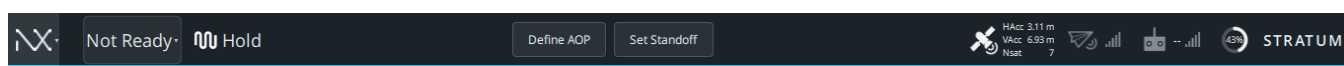
STRATUM's fly view is divided into six regions:

Region	What it does
Top ribbon	Vehicle status, current flight mode, mission shortcuts, GPS accuracy, radio links, battery.
Left command strip (green buttons)	Every action you take on the aircraft -- takeoff, standoff, land, hold, mode selection, abort, engage.
Map (main area)	Live satellite map with the vehicle, the ground station, the trail, the AOP (Area of Persistence), and the standoff ring.
Right camera dock	Record video, take photos, open camera settings.

Region	What it does
Bottom-left video pane	Small live camera feed with telemetry overlay and ROI zoom slider.
Bottom instruments	Telemetry row, artificial horizon (attitude), and compass.

The rest of this guide zooms into each region.

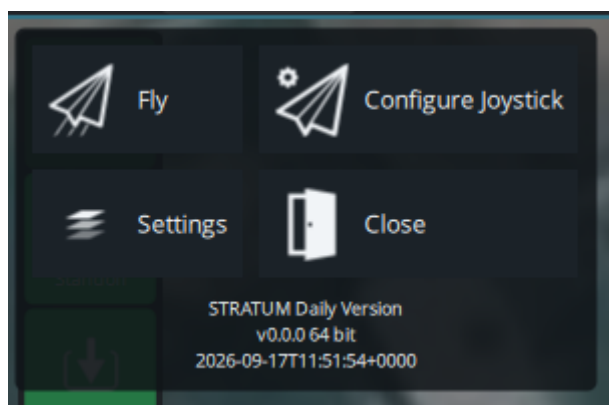
2. Top ribbon -- vehicle status and quick actions



Reading the ribbon left to right:

2.1 NX menu (top-left corner)

Clicking the **NX** logo opens the STRATUM main menu:



- **Fly** -- returns you to the flight screen you see in this guide.
- **Configure Joystick** -- pairs and calibrates a USB/BT joystick or transmitter.
- **Settings** -- application settings: general options, comm links, offline maps, MAVLink, console, mock link, etc.
- **Close** -- closes the STRATUM application.

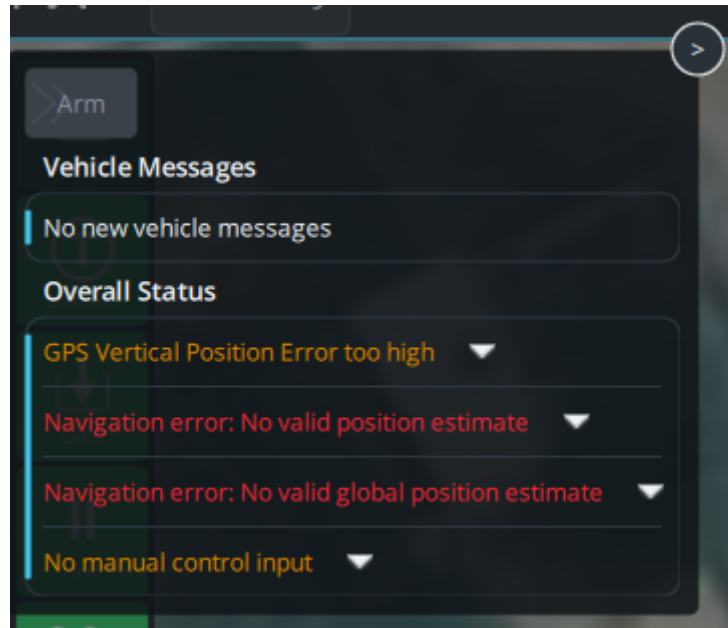
The build version and timestamp are shown underneath (e.g. STRATUM Daily Version v0.0.0 64 bit ...).

2.2 Arm chip -- "Not Ready" / "Ready" / "Armed" / "Flying"

The chip immediately after the NX icon is the **arm status chip**. It shows the current aircraft state in one word and doubles as the entry point into the preflight drawer.

- **Not Ready** -- one or more preflight checks have failed. Click the chip to see the details (Section 2.3).
- **Ready** -- all preflight checks pass. Click the chip to arm the vehicle.
- **Armed** -- motors are spinning / ready to spin. Click to disarm on the ground.
- **Flying** -- the aircraft is in the air.

2.3 Status drawer (opens when you click the arm chip)



The drawer is split into three parts:

1. **Arm** button at the top -- arms the aircraft when all checks pass.
2. **Vehicle Messages** -- the last messages sent by the flight controller (calibration prompts, warnings, informational text).
3. **Overall Status** -- a list of every preflight check the aircraft runs. Any failing check appears in red and blocks arming. Common examples: - *GPS Vertical Position Error too high* -- wait for better GPS lock. - *Navigation error: No valid position estimate* -- the EKF has no valid position; move to open sky. - *No manual control input* -- RC transmitter is off or not paired.

Each row has a chevron you can click to see more detail on that particular check.

2.4 Current flight mode display

The line **^v Hold** (icon + name) shows the mode the flight controller is running **right now**. This is a read-only indicator -- to *change* the mode use the Flight Mode button on the left strip (Section 3.5). On VTOL airframes a second tag appears next to the mode name showing FW / MC / Transition.

2.5 Mission shortcuts -- Define AOP and Set Standoff

Two ribbon buttons let you draw the STRATUM operating geometry directly on the map:

- **Define AOP** -- Area of Persistence. Click, then drag on the map to place the point the aircraft should orbit or watch. This is the orange circle you see in the main screenshot.
- **Set Standoff** -- sets the standoff radius (how far the aircraft stays from the AOP). Click, then drag on the map to size the ring.

2.6 GPS accuracy readout

Three small numbers on the right of the center block:

- **HAcc** -- horizontal position accuracy in metres. Smaller is better; ≤ 1 m indicates a fixed RTK / good GNSS lock.
- **VAcc** -- vertical position accuracy in metres.

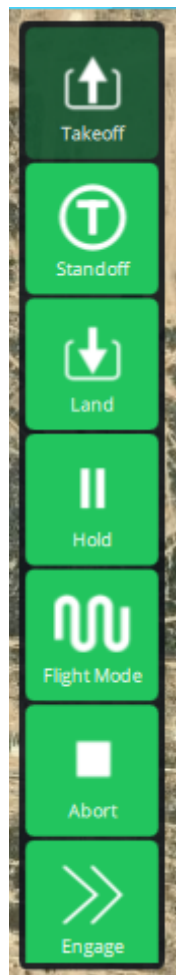
- **Nsat** -- number of GNSS satellites in the solution.

2.7 Link and battery indicators

To the right of GPS, four small icons summarise connectivity and power:

- **Telemetry radio** (antenna icon with bars) -- signal quality of the datalink to the aircraft. Grey bars mean no telemetry.
- **RC controller** (transmitter icon with bars) -- signal quality reported by the RC receiver. Grey bars mean no RC.
- **Battery** (round gauge with %) -- percentage remaining on the primary aircraft battery. Colour changes as the pack drains (green -> amber -> red).
- **STRATUM wordmark** -- brand mark on the right edge; no click action.

3. Left command strip -- the pilot's action buttons



The green strip on the left of the map is the primary way you command the aircraft. Buttons are stacked in the order you typically use them.

3.1 Takeoff

Sends a takeoff command. A slider confirmation appears so you can slide to commit. The aircraft climbs to the takeoff altitude configured in vehicle parameters and then holds position.

3.2 Standoff

Commands the aircraft to fly to the currently defined standoff position (the ring you drew with **Set Standoff** on the top ribbon) and hold there. Use this to reposition the aircraft after retasking the AOP or when returning from a manual dash.

3.3 Land

Commands a land at the current position. Confirm on the slider that appears. The aircraft descends and disarms once it detects touchdown.

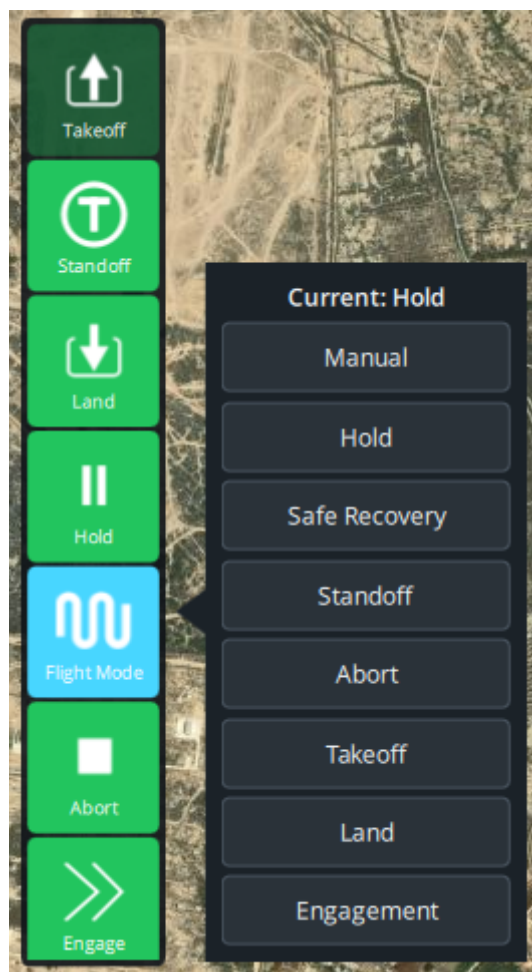
3.4 Hold

Commands the aircraft to stop and hover in place immediately -- the fastest way to freeze the mission if you need a moment to think. Multirotors will loiter at the current lat/lon/alt; fixed-wing airframes will orbit.

3.5 Flight Mode

Opens the mode picker (see Section 3.5.1). Use this button whenever you need to switch the flight controller between modes such as *Manual*, *Hold*, *Standoff*, *Engagement*, etc.

3.5.1 Flight Mode picker



The picker lists every mode STRATUM allows you to command:

- **Current: Hold** -- the header shows the mode the aircraft is in right now.
- **Manual** -- full stick-authority piloting; safety limits are minimal.
- **Hold** -- hover in place at the current position.

- **Safe Recovery** -- return to the recovery point using the safest path configured by the operator.
- **Standoff** -- fly to the standoff ring and orbit at the configured radius.
- **Abort** -- break off the current task cleanly (identical to the **Abort** strip button, offered here for convenience).
- **Takeoff** -- command a takeoff from the ground.
- **Land** -- command a landing at the current position.
- **Engagement** -- commit the aircraft to the engagement profile you configured (used with the **Engage** strip button below).

Tapping any row asks you to confirm the mode change before it is sent to the aircraft.

3.6 Abort

Clears the current tasking and puts the aircraft into a safe holding behaviour immediately. Use this if a mission is going wrong and you need to disengage.

3.7 Engage

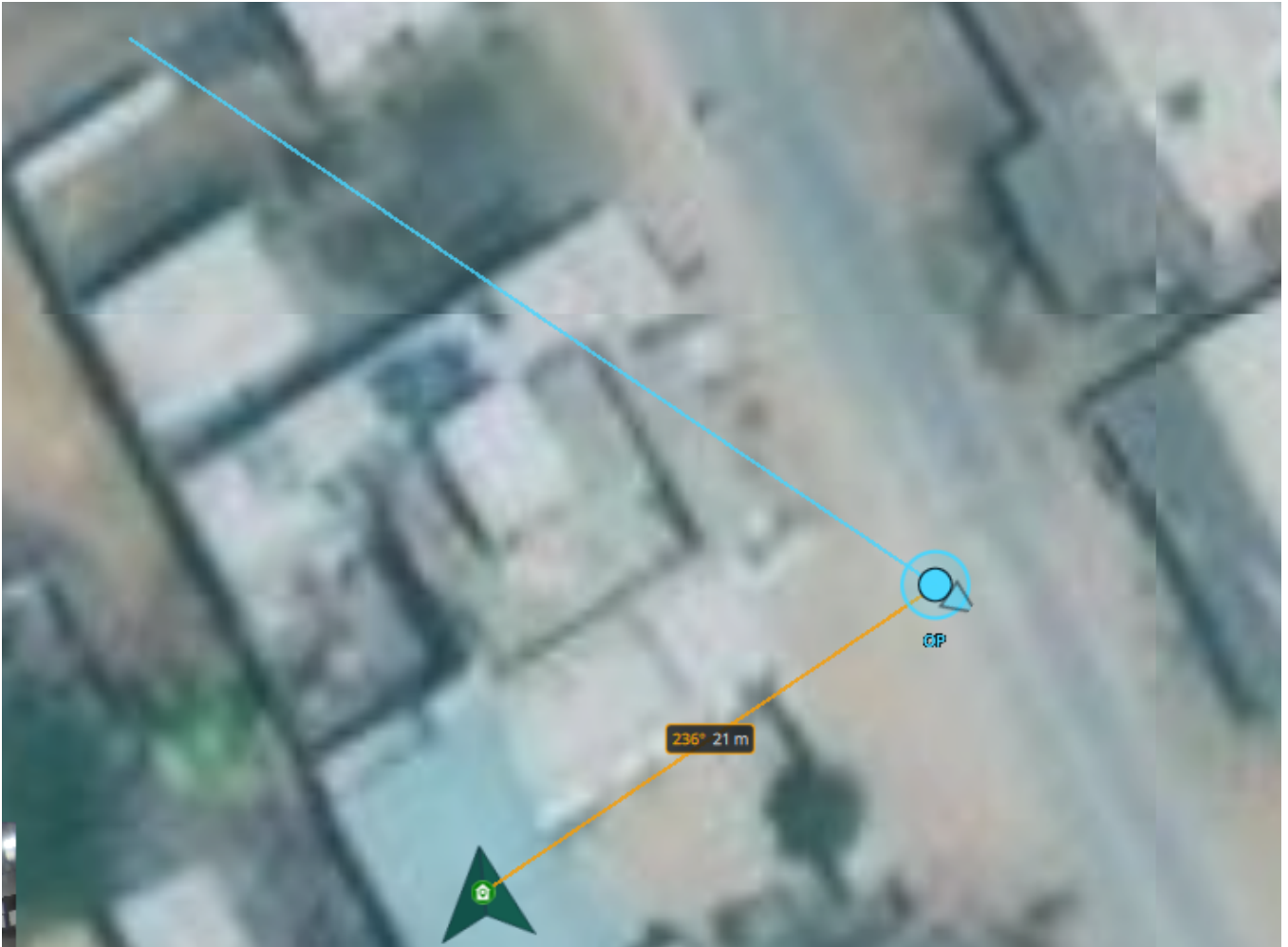
Commits the aircraft to the engagement profile. This is a destructive / irreversible command -- STRATUM will present a red confirmation slider so you cannot trigger it accidentally.

4. Map view -- situational awareness

The main satellite map shows every geometry that matters to the mission:

- **Orange circle** -- the **AOP** (Area of Persistence) you defined from the ribbon. The aircraft's job is anchored to this point.
- **Cyan circle** inside the AOP with an arrow -- the **vehicle**. The arrow shows the aircraft heading. The label next to it shows range and bearing to another reference (`157° 30 m` in the screenshot).
- **Green diamond** -- an optional waypoint or reference target on the map.
- **500 m scale bar** in the top-left of the map -- current zoom scale.

4.1 GCS icon, heading, and trail

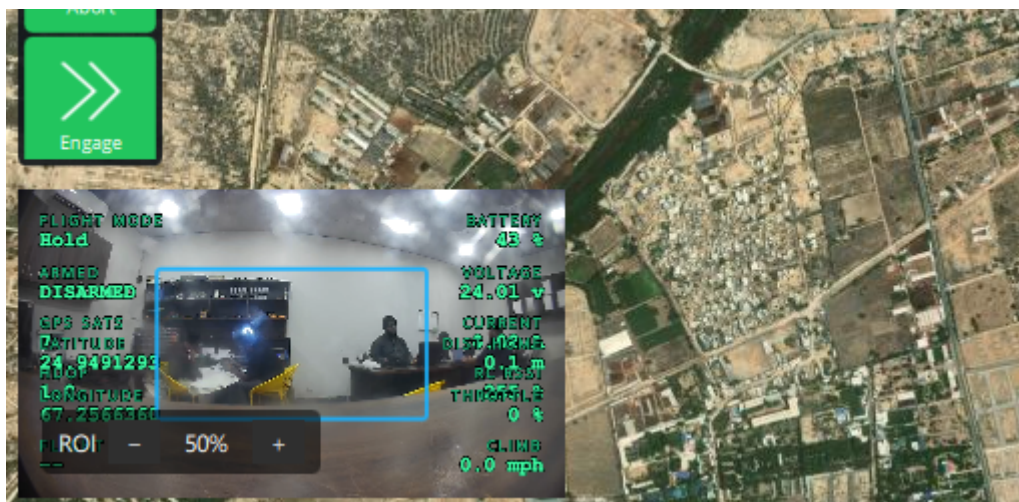


STRATUM draws your **ground station** on the map so you always know where you are relative to the aircraft:

- **Blue circle with a chevron** -- the GCS position. The chevron points in the direction the GCS is currently facing (heading). If the GNSS receiver does not report a heading (stationary receivers usually don't), the chevron falls back to your direction of travel or the bearing to the aircraft and is drawn slightly transparent to indicate it is estimated.
- **GCS label** -- printed under the marker so it is unambiguous.
- **Blue dotted line** -- the GCS **trail**: where the GCS has been over the last few seconds. Useful when moving with the aircraft on foot or in a vehicle.
- **Orange line with 236° 21 m** -- the **bearing and distance from the GCS to the vehicle**. This is your quickest "where is the aircraft relative to me?" cue.

The GCS position comes from the NMEA GNSS receiver you configured under **Settings -> Comm Links** (port and baud rate). If nothing is drawn, check that the NMEA port is opened by STRATUM (and not held by another application like UPrecise).

5. Video window (bottom-left)



The bottom-left corner shows the live camera feed with a compact telemetry overlay burned in. Click the pane to swap it with the map (picture-in-picture style).

Overlay fields (green text):

- **FLIGHT MODE** -- current mode (e.g. `Hold`).
- **ARMED / DISARMED** -- arm state of the vehicle.
- **BATTERY** -- remaining battery percentage.
- **VOLTAGE** -- pack voltage.
- **GPS SATS** -- number of satellites the *aircraft* sees.
- **CURRENT** -- battery current draw in amps.
- **LATITUDE / LONGITUDE** -- aircraft position.
- **HEADING** -- aircraft magnetic heading.
- **ALTITUDE** -- altitude above the takeoff point.
- **THROTTLE** -- throttle percentage.
- **AIRSPEED / GROUND SPEED** -- knots or mph, depending on units settings.

At the bottom of the pane:

- **ROI - 50 % +** -- Region of Interest zoom. The `-` and `+` buttons zoom the digital ROI in the camera image; the percentage shows the current zoom level. This is a digital zoom -- it does not command a physical gimbal focal-length change.

6. Right-hand camera dock

On the right edge of the map (visible in the main screenshot) STRATUM shows camera capture controls:

- **Red circle -- "Video"** -- starts and stops video recording. The counter underneath (e.g. `00:00:00`) shows the current clip length. The circle turns into a stop icon while recording.
 - **Grey circle -- "Photo"** -- captures a still photo. The counter below (e.g. `00000`) shows how many photos you've taken this session.
 - **Gear icon -- Camera settings** -- opens camera options such as source selection, resolution, exposure, and storage location.
-

7. Bottom instruments -- telemetry and attitude

Along the bottom of the fly view you see three groups:

7.1 Telemetry row (center)

A compact grid of numbers with icons:

- **^ 0.7 m** -- climb / descent rate.
- **-> - . - m** -- distance to next waypoint (dashes when unset).
- **Latitude / Longitude** -- aircraft geodetic position.
- **^ 0.0 mph** -- vertical speed in your units.
- **-> 0.0 mph** -- horizontal ground speed.
- **T 00:00:00** -- elapsed flight time since arming.
- **^ 0.0 m** -- altitude AGL / above takeoff.
- **Alt (Rel) 0.7 m** -- altitude relative to the takeoff point.

The exact set of fields is configurable in **Settings -> General -> Instrument Panel**.

7.2 Attitude indicator (artificial horizon)

The round instrument with the blue / green split and the pitch ladder shows the aircraft's **roll and pitch** in real time. The centre wings represent the aircraft; the horizon tilts and slides behind them.

7.3 Compass

The compass rose on the far bottom-right shows:

- **N / E / S / W** cardinal points.
- **Red arrow** -- aircraft heading.
- **Numeric heading readout** underneath (e.g. **354° S**).

8. A typical mission -- putting it all together

1. **Power up** the aircraft and the ground station. Wait for the arm chip to turn from **Not Ready** to **Ready**. If it stays red, click the chip and resolve every red item in the **Overall Status** list.
 2. **Draw your geometry** with **Define AOP** and **Set Standoff** on the ribbon.
 3. **Takeoff** using the left-strip **Takeoff** button. Confirm on the slider.
 4. **Standoff** -- after takeoff the aircraft holds. Press **Standoff** to send it to the ring, or leave it in **Hold** while you verify camera and links.
 5. **Change modes** with **Flight Mode** whenever you need something other than Standoff / Hold (for example, **Manual** for hands-on flying).
 6. **Record video / take photos** with the right-hand dock as needed.
 7. **Engage** only when you have explicitly cleared the mission -- the confirm slider is there specifically to prevent accidents.
 8. **Land** or **Abort** to end the sortie. Watch the status drawer for any post-flight messages before shutting down.
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9. Troubleshooting quick reference

Symptom	First thing to check
Arm chip stays Not Ready	Click the chip, resolve each red row in the Overall Status list.
No GCS marker on the map	Settings -> Comm Links: NMEA port + baud correct; no other app (e.g. UPrecise) is holding the port.
GCS chevron looks faded / stuck	Your GNSS receiver isn't reporting heading; STRATUM is estimating from motion or bearing. Move a few metres and it will settle.
Flight Mode picker opens but rows look empty	Rebuild after the latest STRATUM update -- this was a known QML scoping issue and is fixed in current builds.
Video pane is black	Check the camera source and format under the gear icon on the right dock.
Radio / RC bars are grey	Telemetry link or RC receiver isn't heartbeating; check power, antennas, and pairing.

STRATUM is built on the QGroundControl platform; the fly view is customised for the STRATUM mission workflow (AOP, Standoff, Engagement). If you're familiar with QGroundControl, everything under Settings works the same way.